

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

4
5 Jaeseung Kim¹ Sang Woo Park^{1,2,*}

6 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

7 **2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

8 *Corresponding author: sangwoopark@snu.ac.kr

9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allow each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

Main

Model

We begin by extending previously established models of CI in which a group of players participates in a prediction task, with each player making an individual prediction based on partial observations of the underlying causal factors and the group forming a collective prediction by aggregating these individual predictions [30, 12, 13]. The prediction target (Y) is modeled as a linear combination of M causal factors (x_i), with fixed coefficients (α_i): $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution. Among M different factors, an individual (A) makes prediction by the multiplication of single factor of interest (x_{e_A}) and their own belief (c_A): $Y_A = c_A x_{e_A}$. Here, we assume each individual can independently consult AI and revise their prediction based on their reliance on AI (β_A): $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$, and the revised prediction is aggregated to form collective prediction.

To model AI, we consider a scenario where the group of players is given a novel prediction task, whereas the AI has an ability to perform prediction based on causal factors because it already been trained on similar tasks. Every time a player consults AI, it will make predictions (Y_A^{AI}) based on its internal parameters, which can include factor-specific bias (b_i) and stochastic error (ϵ). We then consider two extreme types of AI, which differ in the amount of information AI uses when making predictions. At one extreme, the AI operates with complete knowledge of all causal factors, using them regardless of the user's interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. At the other extreme, AI can work like a chatbot, in which AI predicts the outcome based only on the user's interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’ and ‘Chatbot AI’, respectively.

Each generation consists of multiple round of predictions, and each player is rewarded by a payoff according to the incentive structure. For simplicity, we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff rather than computing the payoff for each round. We consider two incentive structures, “feedback” and “niche-expert” [13], which have been shown to promote near-perfect collective accuracy in the absence of AI (except averaging aggregation with niche-expert structure). The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy. In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a underrepresented interest.

Through time, the group evolves by imitating the peers who receive higher payoffs. Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (factor of interest, belief, reliance on AI) of the other with a probability determined by the difference between their expected payoffs. Individual predictions are aggregated into a collective prediction ($\hat{Y}$) by simple averaging when

103 using Omniscient AI and by clustering when using Chatbot AI. Collective accuracy
 104 is defined as the reduction in prediction error achieved by the collective prediction
 105 relative to a baseline prediction (α_0). Hereafter, we refer to the near-perfect collective
 106 accuracy as accurate CI.

107 This model now allows us to explore the impact of AI on CI depending on the
 108 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 109 of CI with and without AI and subsequently investigate a better incentive structure
 110 that can promote CI in the presence of AI.

111 **AI can either promote or prevent the emergence of CI**

112 To understand the impact of AI, we first compare collective dynamics with and
 113 without AI. We find that, except when Omniscient AI is used under the feedback
 114 structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect
 115 of AI differs depending on the type of AI.

116 When Omniscient AI assists players, the group becomes equally or more accurate
 117 than the group without AI. Under the feedback structure, a group achieves accurate
 118 CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert
 119 structure, the group without AI is no better than the baseline prediction. Although
 120 Omniscient AI cannot help players achieve accurate CI, the group’s prediction with
 121 AI is more accurate than both that of the group without AI and the AI itself (Fig.
 122 2B). These results are robust across different levels of AI accuracy and can be shown
 123 mathematically by assuming a sufficiently large group of players (Fig. S2).

124 When Chatbot AI assists players, the group becomes less accurate than the group
 125 without AI. Although both incentive structures can achieve accurate CI without AI,
 126 the assistance of Chatbot AI hinders group performance and prevents accurate CI
 127 from emerging (Fig. 2C, D). This occurs through different mechanisms depending
 128 on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss
 129 of interest diversity, which eventually makes it impossible for the group to account
 130 for variation in some causal factors. This leads to an increase in collective variance
 131 and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity
 132 may arise even without AI after sufficiently many generations in a finite population.
 133 However, Chatbot AI makes the group more vulnerable to the diversity loss. Un-
 134 der the niche-expert structure, the group becomes overly reliant on AI, causing the
 135 collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in
 136 collective bias and a decrease in collective accuracy. The niche-expert structure may
 137 also drive reliance on AI toward zero, depending on the relationship between the
 138 true coefficient and AI bias (Fig. S2). In this case, the group completely excludes
 139 AI assistance and builds CI on its own, which substantially delays the emergence of
 140 accurate CI.

141 To investigate whether the group can retain its predictive ability when AI as-
 142 sistance suddenly becomes unavailable at a given generation, we measure counter-
 143 factual collective accuracy. This measure follows the same evolutionary trajectory

144 as the simulation with AI, but accuracy is computed using the players’ unrevised
 145 predictions. We find that counterfactual collective accuracy is always lower than the
 146 collective accuracy of groups, regardless of the AI mode or incentive structure.

147 In summary, AI assistance may promote CI when it is omniscient, whereas AI
 148 operating with limited information may prevent the emergence of accurate CI. These
 149 findings underscore the need for incentive structures that can mitigate the potential
 150 risks associated with AI assistance.

151 **Directly incentivizing or penalizing AI use can additionally** 152 **harm CI**

153 Given the strong influence of AI on collective outcomes, a straightforward interven-
 154 tion is to directly incentivize or penalize the use of AI. To investigate whether such
 155 direct incentives or penalties help emergence of CI, we modify the two incentive
 156 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

157 where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

158 Directly incentivizing or penalizing AI use further reduce the collective accuracy
 159 (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure
 160 (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to
 161 always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some
 162 cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can
 163 achieve higher collective accuracy than the original incentive structure. However, in
 164 most cases, penalizing AI use imposes strong selection against AI reliance, pushing
 165 the group toward lower stationary collective accuracy. Although the group with
 166 Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn’t
 167 result in stationary collective accuracy, incentivizing or penalizing AI use further
 168 strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing
 169 or penalizing the AI use may exacerbate rather than resolving the problem of AI.

170 **Balanced incentives mitigate AI-induced failures of CI**

171 Neither the original incentive structures nor simple interventions targeting AI use
 172 are sufficient to promote CI. What incentive structure, then, would work in the
 173 presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures
 174 fail to promote accurate CI, we investigate whether a better structure can overcome
 175 the risks introduced by AI. Here, we propose a “balanced” structure, defined as the
 176 simple average of the feedback and niche expert incentives.

177 Core idea behind the balanced structure is to maintain interest diversity while
 178 preventing the collective bias of the group. The feedback structure is prone to in-
 179 terest diversity loss, whereas can strongly prevent collective bias by incentivizing
 180 players whose predictions lie opposite to the collective prediction. By contrast, the

181 niche-expert structure can maintain interest diversity because it directly incentivizes
182 minority interests. However it cannot prevent collective bias because, within the
183 same factor of interest, individuals are rewarded only according to their individual
184 predictive performance, which homogenize the predictions.

185 By integrating both structure, the balanced structure can promote accurate CI by
186 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-
187 pare to the feedback and niche-expert, players are both evenly distributed through
188 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-
189 tual collective accuracy maintained high compare to both feedback and niche-expert.
190 These patterns are driven by the emergent property of the balanced structure, in
191 which group increases its reliance on AI as the AI becomes more accurate, demon-
192 strating appropriately calibrated AI use.

193 Discussion

194 In this paper, we study how AI and incentive structures affect the emergence of CI.
195 Our model suggests that the effect of AI depends on the type of AI used by the
196 group. AI that can incorporate all relevant causal factors may improve group perfor-
197 mance, whereas AI that relies on users to gather information about the causal factors
198 may impair it. By examining different variants of incentive structures, we further
199 show that direct interventions targeting AI use generally harm group performance.
200 Instead, we suggest balanced structure which can promote CI while also generating
201 appropriately calibrated AI use as an emergent property.

202 Although Omniscient AI can produce group performance that is equal to or bet-
203 ter than that achieved without AI, such AI may be difficult to realize in practice
204 because it needs to be able to observe causal factors independently. Achieving this
205 capability would require hardwares, such as sensors, as well as autonomous data ac-
206 quisition capabilities. For this reason, most AI services currently available are better
207 represented by the Chatbot AI in our model, which highlights the potential risks of
208 AI use for CI.

209 In our model, individuals revise their predictions based on the AI’s assistance. It
210 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
211 dictions, a pattern that has also been observed empirically. However, a reduction in
212 prediction diversity does not necessarily lead to the collapse of CI. The principal risk
213 arises when the incentive structure fails to guide the group toward an appropriate
214 set of strategies. The feedback structure leads to the breakdown of CI because it
215 recruits more players to more important factors, which have larger coefficients. This
216 characteristic creates a risk of interest-diversity loss in finite populations, a situa-
217 tion in which no player remains interested in certain factors because most players
218 shift toward relatively important factors. Without AI, this occurs when initial in-
219 dividual predictions are not sufficiently diverse or when there is no source of novel
220 ideas, such as mutation. Revising individual predictions with AI assistance exac-

221 erbates this problem by pulling individuals’ predictions toward the AI’s prediction,
222 thereby reducing prediction diversity even under the same belief distribution. The
223 niche-expert structure hinders CI because it favors extreme levels of reliance on AI.
224 Because individuals within each interest group are affected only by their own perfor-
225 mance, incentives based on individual performance reach their maximum at either
226 full reliance on AI or no reliance at all. This extreme reliance results in suboptimal
227 collective accuracy or substantially delays the emergence of CI.

228 Because AI already generates substantial benefits as well as costs, organizations
229 increasingly seek to improve collective performance by regulating AI use. Such in-
230 terventions may take direct forms, including AI-use policies or token-based leader-
231 boards, or may operate indirectly through other incentives, such as productivity
232 rewards or the social stigma associated with AI use. However, our model suggests
233 that directly incentivizing or penalizing AI use is unlikely to induce appropriately
234 calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overre-
235 liance, causing groups to propagate AI biases, whereas penalizing AI use could lead
236 to underreliance, preventing groups from using AI even when it performs well in the
237 relevant domain. Our result instead shows that appropriate AI use and accurate CI
238 can be promoted by focusing on the classic properties of groups that have been em-
239 phasized for decades—maintaining diversity and preventing collective bias—rather
240 than on AI use itself.

241 The consistently limited counterfactual collective accuracy observed across AI
242 types and incentive structures suggests that the presence of AI can degrade hu-
243 man group’s own performance, consistent with empirical evidence. This may reduce
244 collective performance when AI assistance becomes unavailable because of resource
245 constraints, system failure, or model collapse, forcing the group to rebuild CI through
246 further adaptation. The higher counterfactual collective accuracy achieved under the
247 balanced structure indicates that an appropriate incentive structure can mitigate the
248 risk of CI breakdown following the loss of AI assistance.

249 Our model and analysis have several limitations. First, the model considers only
250 social learning among humans and treats the influence of AI as unidirectional. In
251 reality, humans can learn from AI, while AI systems can also be further trained
252 on human-generated data. Such bidirectional adaptation may further reduce group
253 performance by increasing the similarity between human and AI predictions and
254 thereby weakening their complementarity. Empirical studies of human–AI collabo-
255 ration likewise suggest that stronger correlations between human and AI predictions
256 can diminish the benefits of collaboration. Second, we model AI errors using factor-
257 specific additive biases and random error. Previous studies of AI bias and error
258 suggest that additive biases may arise from skewed training data, or systematic cal-
259 ibration. However, AI errors may also exhibit more complex structures, including
260 biased estimates of factor effects and correlations across prediction errors. Further
261 empirical research is therefore needed to characterize the structure of AI bias and
262 error, together with modeling approaches that can incorporate these more complex
263 patterns. Finally, to retain the tractability of established CI models, we assume that

the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population N consisting of n players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and predicts the target by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. Although the bias is indexed by factor, all b_i , including b_0 , are assigned the same value in the main analyses for simplicity. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore held τ constant and varied the magnitude of the additive bias to manipulate the level of AI accuracy across analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of AI used by the group.

When the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{n}. \quad (7)$$

327 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 328 and the AI’s prediction are based on the same causal factor. Each revised predic-
 329 tion therefore contains information about only one factor and can be treated as an
 330 estimate of that factor’s contribution to the target. We consequently use clustering
 331 aggregation.

332 Clustering aggregation is performed by grouping individuals according to their
 333 interests and calculating the mean prediction within each group. The collective
 334 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

335 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 336 of players in that set.

337 When the group is not assisted by AI, collective predictions are computed from the
 338 unrevised individual predictions throughout the simulation. Both simple averaging
 339 and clustering can be used in the absence of AI. For fair comparison, we use the
 340 same aggregation rule as in the corresponding AI-assisted condition.

341 Incentive structures

342 Players receive payoffs determined by an incentive structure based on their strategies.
 343 We consider five incentive structures: feedback structure, niche-expert structure,
 344 direct intervention on AI use of these two structures, and a balanced structure.

345 The feedback structure does not directly reward individuals according to their
 346 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 347 residual error of the collective prediction and therefore have the potential to move
 348 the collective prediction toward the true value. Given player A ’s revised prediction
 349 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

350 The niche-expert structure assigns higher payoffs to individuals who make accu-
 351 rate predictions while having an underrepresented interest. Specifically, the squared
 352 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 353 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

354 Previous research has shown that these two incentive structures can produce
 355 nearly perfect CI through social learning, except when the niche-expert incentive is
 356 combined with simple averaging. They are therefore used as baseline structures for
 357 examining how AI assistance affects the emergence of CI.

358 To examine the effects of directly encouraging or discouraging AI use, we mod-
 359 ify the feedback and niche-expert structures by adding a term proportional to the
 360 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

361 Here, λ is a weighting coefficient representing the strength of the direct incentive
 362 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 363 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 364 and the payoff functions reduce to the original feedback and niche-expert structures.

365 The balanced structure is defined as the arithmetic mean of the feedback and
 366 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

367 Simulations

368 All simulations were conducted using an agent-based model with 50 causal factors
 369 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 370 target were sampled at the beginning of the simulations and subsequently held fixed.

371 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 372 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 373 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

374 Players’ initial strategies were also sampled from predefined probability distri-
 375 butions. The distribution of individual belief c_A depended on the aggregation rule.
 376 Under simple averaging, each factor’s contribution is scaled by the proportion of
 377 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 378 tribution with mean zero and standard deviation 100, following the scaling used
 379 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 380 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 381 are averaged separately within each interest group and then summed, no additional
 382 scaling of c_A is required. The initial interest of each player was sampled uniformly
 383 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 384 distribution on $[0, 1]$.

385 We assumed that prediction events occurred on a substantially faster timescale
 386 than social learning. At each generation, we therefore calculated the expected payoff
 387 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
 388 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
 389 and the stochastic AI error ϵ , conditional on the current population strategies and
 390 the fixed target parameters. Closed form expression of expected payoffs were derived
 391 for all incentive structures and are presented in the Supplementary Methods.

Each generation consisted of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , were randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduced a mutation rate μ , such that with probability μ , player A did not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate was set to zero in all simulations except those presented in Supplementary Figure 4.

Unless otherwise specified, each simulation was run for 200,000 generations. When a result was represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we defined six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances were evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy was normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

References

- [1] Francis Galton. Vox populi, 1907.
- [2] Anita Williams Woolley, Christopher F Chabris, Alex Pentland, Nada Hashmi, and Thomas W Malone. Evidence for a collective intelligence factor in the performance of human groups. *science*, 330(6004):686–688, 2010.
- [3] Ilan Yaniv and Maxim Milyavsky. Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes*, 103(1):104–120, 2007.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

- [4] Cars Hommes, Joep Sonnemans, Jan Tuinstra, and Henk Van de Velden. Coordination of expectations in asset pricing experiments. *The Review of Financial Studies*, 18(3):955–980, 2005.
- [5] Irving Lorge, David Fox, Joel Davitz, and Marlin Brenner. A survey of studies contrasting the quality of group performance and individual performance, 1920–1957. *Psychological bulletin*, 55(6):337, 1958.
- [6] Tabea Dreyer, Amir Haluts, Amos Korman, Nir Gov, Ehud Fonio, and Ofer Feinerman. Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, 122(1):e2414274121, 2025.
- [7] Andrew M Simons. Many wrongs: the advantage of group navigation. *Trends in ecology & evolution*, 19(9):453–455, 2004.
- [8] Andrew Berdahl, Colin J Torney, Christos C Ioannou, Jolyon J Faria, and Iain D Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013.
- [9] Christoph Riedl, Young Ji Kim, Pranav Gupta, Thomas W Malone, and Anita Williams Woolley. Quantifying collective intelligence in human groups. *Proceedings of the National Academy of Sciences*, 118(21):e2005737118, 2021.
- [10] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [11] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [12] Richard P Mann and Dirk Helbing. Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences*, 114(20):5077–5082, 2017.
- [13] Guocheng Wang, Qi Su, Long Wang, and Joshua B Plotkin. Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences*, 122(51):e2516535122, 2025.
- [14] Justin Werfel. Fluid thinking about collective intelligence. *Nature Machine Intelligence*, 8(4):506–516, 2026.
- [15] Hao Cui and Taha Yasseri. AI-enhanced collective intelligence. *Patterns*, 5(11):101074, 2024.
- [16] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651(8107):914–919, 2026.
- [17] Anil R Doshi and Oliver P Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.

- [18] Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarlaschi, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, et al. Artificial intelligence index report 2026. *arXiv preprint arXiv:2606.15708*, 2026.
- [19] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12):2293–2303, 2024.
- [20] Jason W Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A Bakker, Joshua A Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, et al. How large language models can reshape collective intelligence. *Nature human behaviour*, 8(9):1643–1655, 2024.
- [21] Lennart Meincke, Gideon Nave, and Christian Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature human behaviour*, 9(6):1107–1109, 2025.
- [22] Andreas Fügener, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. Will humans-in-the-loop become borgs? merits and pitfalls of working with AI. *MIS quarterly*, 45(3):1527–1556, 2021.
- [23] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- [24] Lucy M Aplin, Damien R Farine, Richard P Mann, and Ben C Sheldon. Individual-level personality influences social foraging and collective behaviour in wild birds. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789):20141016, 2014.
- [25] Lu Hong and Scott E Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.
- [26] Isabelle Bousquette. Why some companies say AI ‘tokenmaxxing’ is key to survival. <https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128>, April 2026. The Wall Street Journal. Accessed: 2026-07-07.
- [27] Tim Keary. Is the cult of ‘tokenmaxxing’ just another fad or the new normal? <https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxing-just-another-fad-or-the-new-normal/>, April 2026. Forbes. Accessed: 2026-07-07.
- [28] Dalmeet Singh Chawla. Researchers who use hallucinated references to face arXiv ban. *Nature*, 653(8116):988–989, 2026.

- 497 [29] H Holden Thorp. Chatgpt is fun, but not an author. *Science*, 379(6630):313–313,
498 2023.
- 499 [30] Lu Hong, Scott E Page, and Maria Riolo. Incentives, information, and emergent
500 collective accuracy. *Managerial and Decision Economics*, 33(5-6):323–334, 2012.

501 Acknowledgements

502 S.W.P. was supported by the New Faculty Startup Fund from Seoul National Univer-
503 sity, and the Global-LAMP Program of the National Research Foundation of Korea
504 (NRF) grant funded by the Ministry of Education (No. RS-2023-00301976).

505 Contributions

506 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
507 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
508 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
509 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

510 Competing Interests

511 The authors declare no competing interests.

512 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ($n=8000$ posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$ posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$. (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from $n=8000$ posterior samples. Shaded regions represent the 95% credible interval in our estimates ($n=8000$ posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.